

IPW3 - Polytechnique Montréal Results



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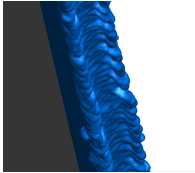
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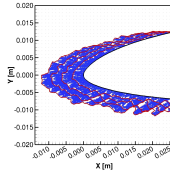
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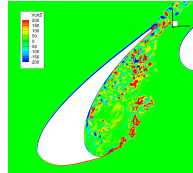
CHAMPS – Overview & Applications



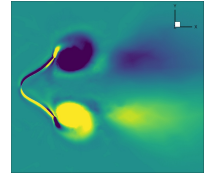
Deterministic Icing



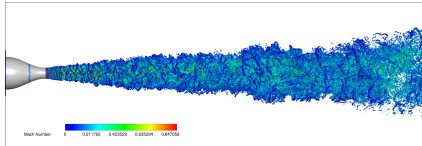
Stochastic Icing



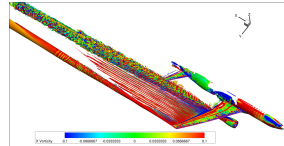
Scale-Resolving CFD (DDES)



Fluid-Structure Interaction



Contrails*



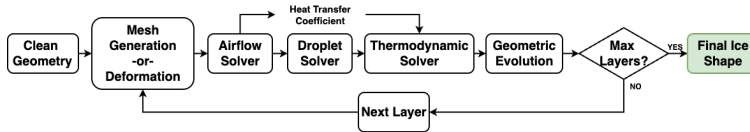
CHAMPS – Chapel MultiPhysics Software

In-house unstructured finite-volume framework developed at Polytechnique Montréal in Chapel for high-performance, multi-fidelity and multiphysics aerospace simulations, providing a unified computational framework for all simulation capabilities.

* Contrail modeling developed in collaboration with Prof. Roberto Paoli.

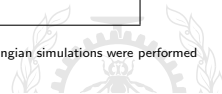


CHAMPS – Icing Framework



Model / Method	NACA0012	ONERA M6
Turbulence Model	Spalart–Allmaras (SA)	
Heat-Transfer Method	One Solution + T_{rec}	
Droplet Model	Eulerian / Lagrangian	
Droplet Drag Model	Deformable Sphere	
SLD Corrections	None	
Thermodynamic Model	Messinger	
Ice Density	580 kg/m ³	680 kg/m ³
Surface Roughness	0.2667 mm / 0.5334 mm	1.0 mm
Geometry Evolution	Algebraic (Single-shot) / Level Set (Multilayer)	
Accretion Strategy	Single-shot & 5 Layers	

* Flow equations use Roe convective fluxes with the Venkatakrishnan limiter; Eulerian droplet equations use an upwind scheme with the Van Albada limiter. Lagrangian simulations were performed to verify the Eulerian collection-efficiency predictions.



CHAMPS – Multilayer Ice-Accretion Approach

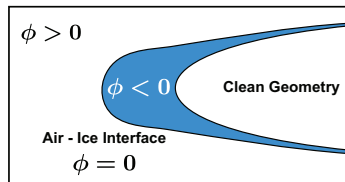
Level-Set Formulation

The ice–air interface is implicitly represented by the **zero level set** of a signed-distance function ϕ [5, 2, 3]:

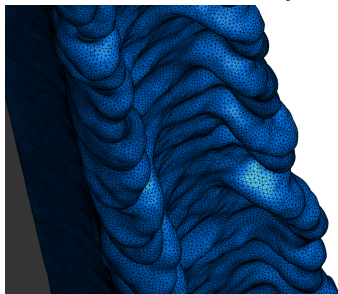
$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{V}) - \phi \nabla \cdot \mathbf{V} = 0, \quad \mathbf{V} = \frac{h_{\text{ice}}}{\Delta t_{\text{layer}}} \mathbf{n}_{\phi_0}.$$

- $\mathbf{n}_{\phi_0}$ denotes the local level-set normal computed from the initial wall-distance field ϕ_0 .
- The predicted ice thickness defines the interface advection velocity $\mathbf{V}$.
- The interface is advected through unsteady time stepping until either the prescribed icing time or the predicted ice volume is reached.
- The updated ice geometry is extracted from the $\phi = 0$ isosurface using the open-source Mmg library [1], and used for the next accretion layer.

Ice–Air Interface



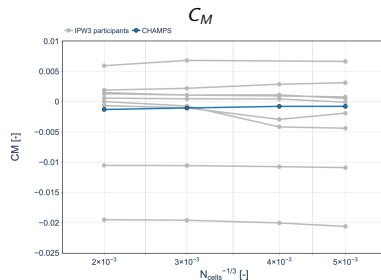
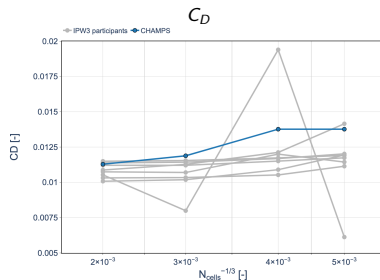
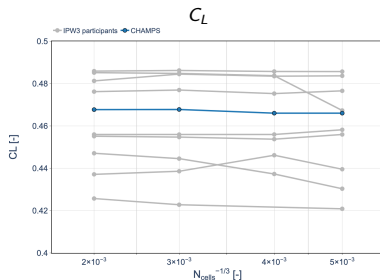
Extracted 3D Ice Geometry



NACA0012 RESULTS



NACA0012 – Integrated Aerodynamic Coefficients

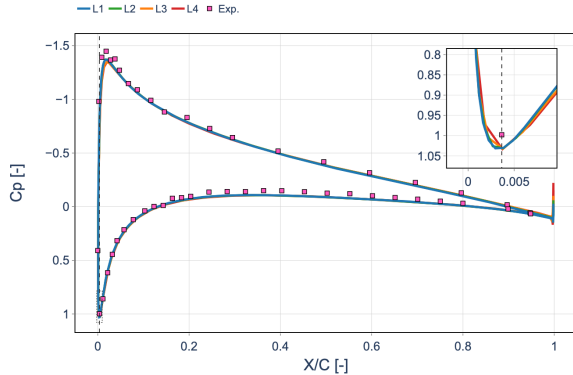
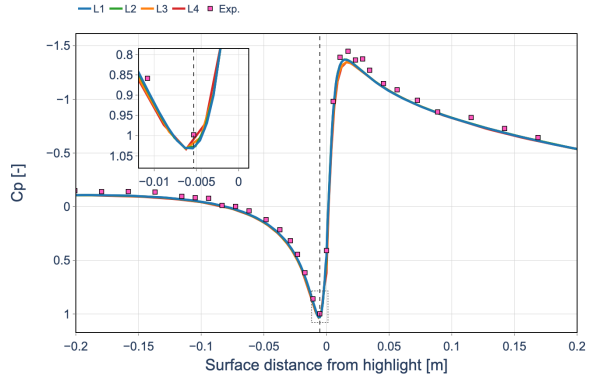


	C_L	C_D	C_M
CHAMPS	0.4677	0.01129	-0.00128
IPW3 L1 Median	0.4657	0.01088	-0.00295
Difference	+0.43%	+3.77%	+0.00167

Integrated Aerodynamic Coefficients Assessment

CHAMPS predicts integrated aerodynamic coefficients close to the IPW3 L1 median. C_D exhibits the greatest sensitivity to spatial grid refinement.

NACA0012 – Pressure-Coefficient Distribution

 C_p vs. x  C_p vs. s 

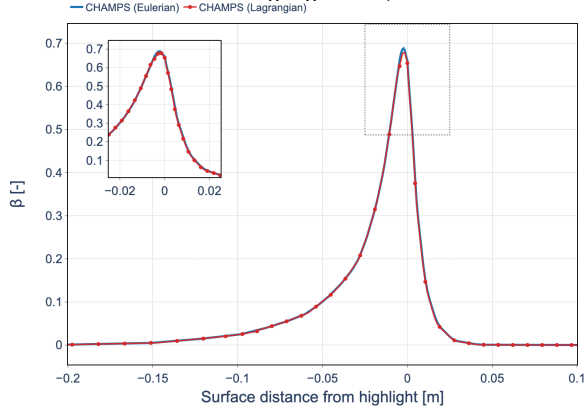
Pressure-Coefficient Assessment

CHAMPS predictions are in good agreement with the experimental pressure distribution across all grid levels.

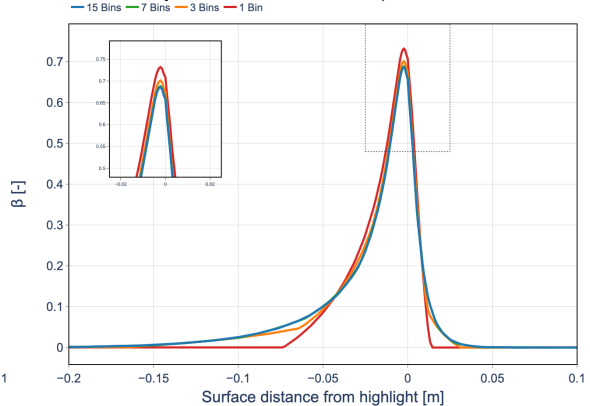


NACA0012 – Droplet Impingement

Eulerian vs. Lagrangian – L1, 7 Bins



Droplet-Bin Refinement – L1, Eulerian

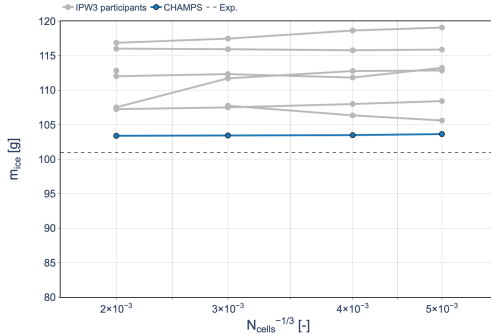


Impingement Assessment

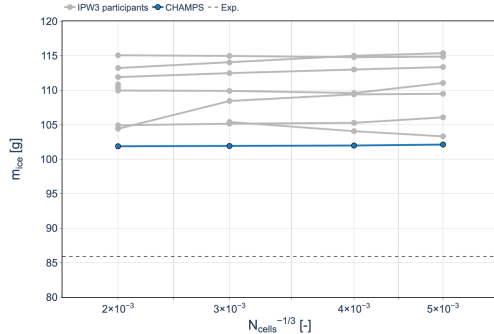
- Close agreement between the CHAMPS Eulerian and Lagrangian formulations, with minor differences near peak collection efficiency.
- Droplet-bin refinement highly affects the peak collection efficiency and impingement limits.

NACA0012 – Ice-Mass Prediction (15 Bins)

AE3932



AE3933

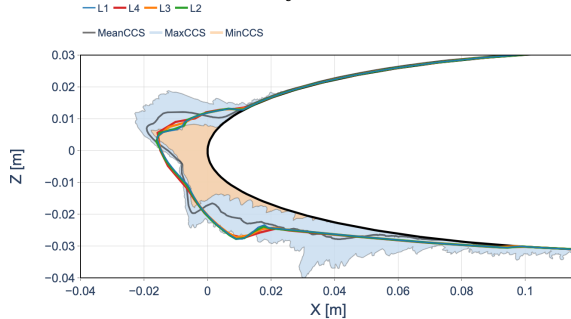


Ice-Accretion Assessment

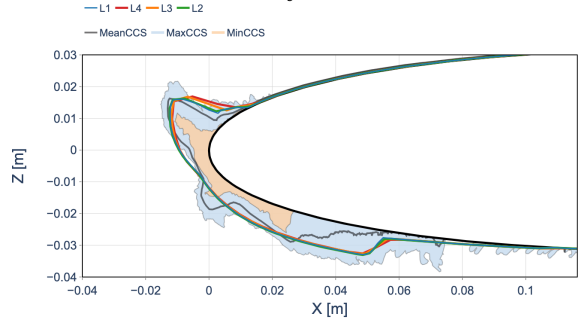
- With the 15-bin distribution, CHAMPS predicts the AE3932 ice mass in very good agreement with the experimental value and exhibits limited sensitivity to spatial grid refinement.
- CHAMPS captures the lower ice mass for AE3933, with a reduction of approximately 1.5 g ($\sim 1.5\%$) relative to AE3932, compared with the approximately 15% reduction measured experimentally.
- Approximately 88–90% of the impinging water mass freezes into ice, with the remaining water mass lost through evaporation.

NACA0012 – Single-Layer Ice-Shape Comparison

AE3932 - $K_s = 0.5334\text{mm}$



AE3933 - $K_s = 0.5334\text{mm}$



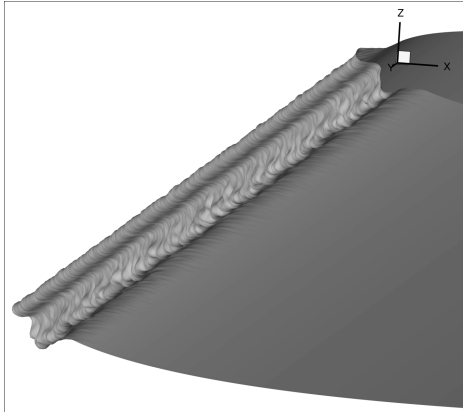
Single-Layer Ice Shapes Assessment

- CHAMPS predicts consistent ice shapes across the grid hierarchy, with limited sensitivity to spatial refinement.
- Predictions remain largely within the experimental ice-shape envelope, with the main differences concentrated around the horn regions.
- Unlike the ice mass, the ice shape captures the effect of the warmer AE3933 condition, with a clear change in upper-horn orientation.

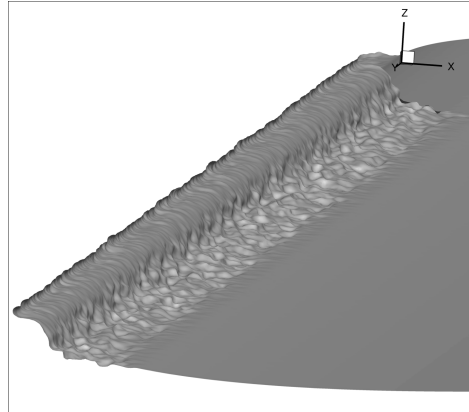


NACA0012 – Multilayer Ice Shapes

AE3932 - $K_s = 0.2667\text{mm}$



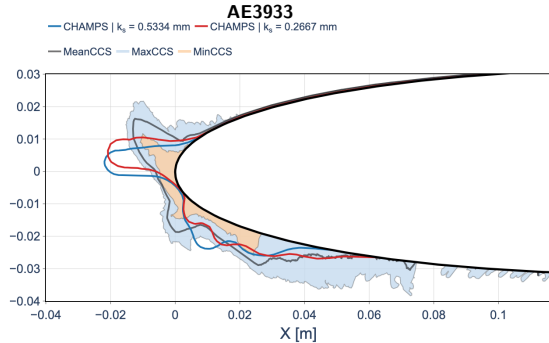
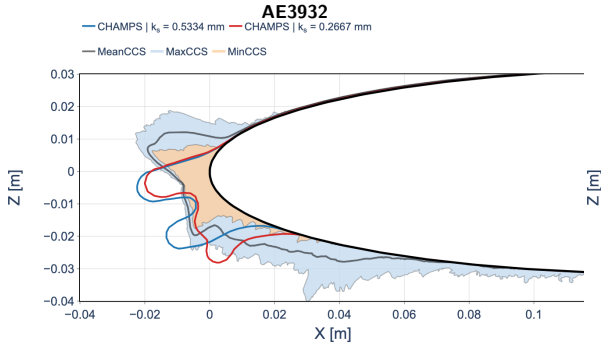
AE3933 - $K_s = 0.2667\text{mm}$



Multilayer Ice Shapes Assessment

- The multilayer simulations capture the spanwise development of the ice horns.
- Distinct three-dimensional ice morphologies are obtained for the two icing conditions, consistent with the differences observed in the two-dimensional cross-sections.

NACA0012 – Multilayer Ice Shapes: Roughness Sensitivity



Multilayer Ice Shapes Assessment

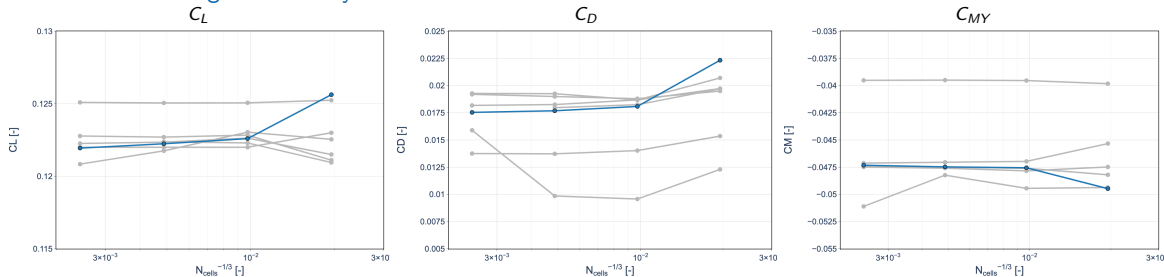
- Surface roughness has a pronounced influence on the multilayer ice shapes, particularly on horn development.
- The lower roughness height provides better overall agreement with the experimental cross sections, but still underpredicts the upper-horn angle.
- The present simulations assume a constant roughness height and constant ice density, which influences the predicted location and development of the horns. Future work will investigate more advanced roughness and ice-density models for these cases.



ONERA M6 RESULTS



ONERA M6 – Integrated Aerodynamic Coefficients

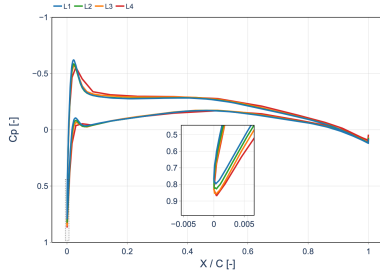
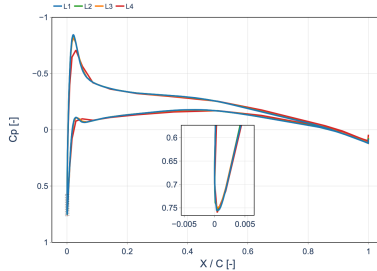
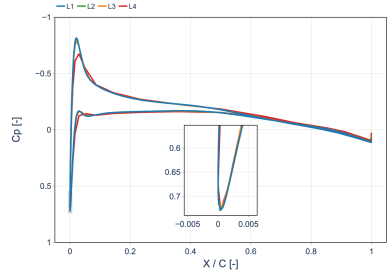


	C_L	C_D	C_{MY}
CHAMPS	0.12194	0.01754	-0.04732
IPW3 L1 Median	0.12213	0.01786	-0.04736
Difference	-0.15%	-1.78%	-0.09%

Integrated Aerodynamic Coefficients Assessment

CHAMPS predicts the L1 integrated aerodynamic coefficients very close to the IPW3 medians, with differences below 2% for all three coefficients. Limited grid sensitivity is observed on the three finest grids, while the coarsest L4 grid produces a more noticeable deviation, particularly for C_D .

ONERA M6 – Pressure-Coefficient Distributions

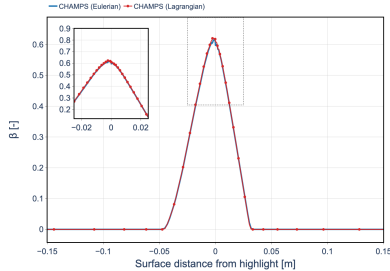
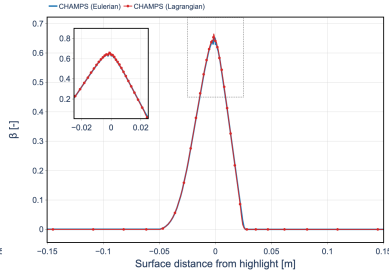
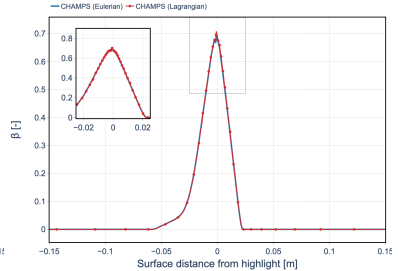
 $y = 0.10 \text{ m}$  $y = 0.75 \text{ m}$  $y = 1.40 \text{ m}$ 

Pressure-Coefficient Assessment

CHAMPS pressure distributions are consistent across the grid hierarchy at all three spanwise stations, with L4 exhibiting the largest deviations.



ONERA M6 – Droplet Collection Efficiency (L1, 1 Bin – Eulerian vs Lagrangian)

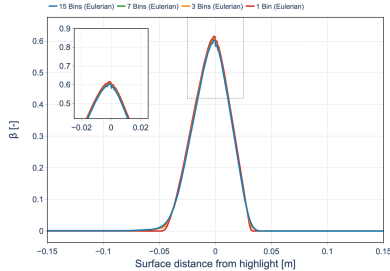
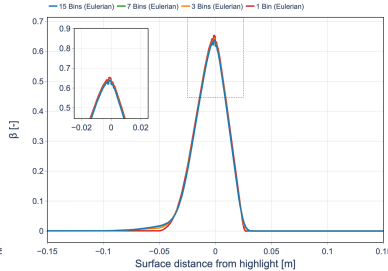
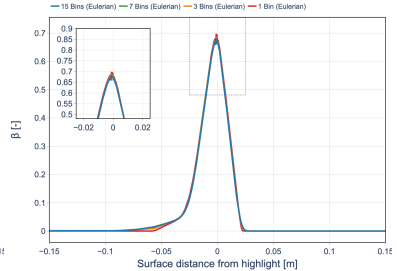
 $y = 0.10 \text{ m}$  $y = 0.75 \text{ m}$  $y = 1.40 \text{ m}$ 

Impingement Assessment

The Eulerian and Lagrangian CHAMPS formulations predict nearly identical collection-efficiency distributions at all three spanwise stations, with consistent peak collection efficiencies and impingement extents.



ONERA M6 – Droplet-Bin Refinement (L1 – Eulerian)

 $y = 0.10 \text{ m}$  $y = 0.75 \text{ m}$  $y = 1.40 \text{ m}$ 

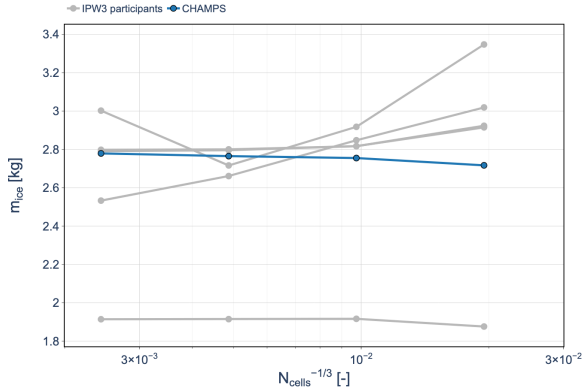
Impingement Assessment

- Droplet-bin refinement has a limited influence on the collection-efficiency distributions at all three spanwise stations.
- Differences between the droplet distributions are less apparent than for the NACA0012 case, as the droplet-size distributions are more concentrated around the MVD.
- The remaining differences are primarily observed near the peak collection efficiency and at the impingement limits.

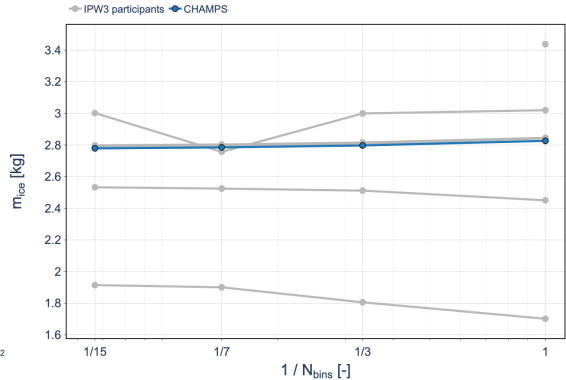


ONERA M6 – Ice-Mass Prediction

Grid Refinement – 15 Bins



Droplet-Bin Refinement – L1

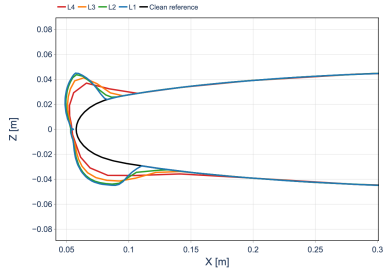
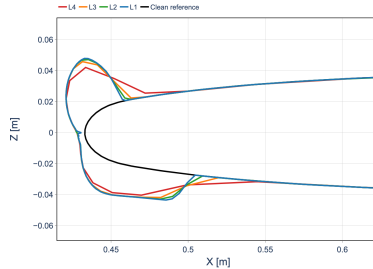
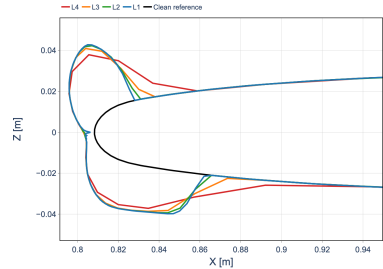


Ice-Accretion Assessment

- CHAMPS predicts approximately 2.8 kg of ice, with limited sensitivity to both spatial grid refinement and the number of droplet bins.
- Approximately 89% of the impinging water mass is converted to ice; the remaining mass is predominantly lost through evaporation.



ONERA M6 – Single-Layer Ice-Shape Comparison

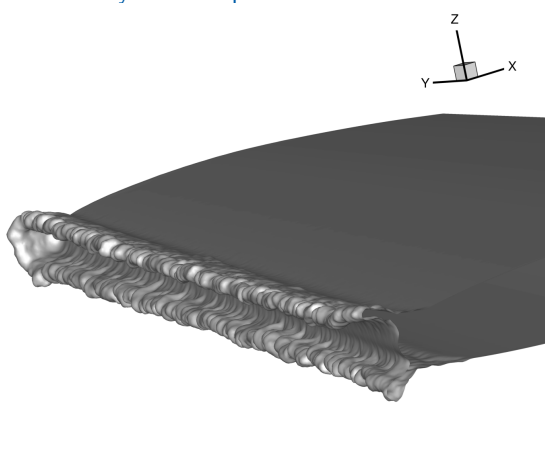
 $y = 0.10 \text{ m}$  $y = 0.75 \text{ m}$  $y = 1.40 \text{ m}$ 

Single-Layer Ice Shapes Assessment

- The single-layer ice shapes show limited sensitivity to spatial grid refinement, with L1–L3 providing similar predictions and the largest deviations observed on L4.
- The larger apparent impingement extent on L4 results from its coarser surface discretization rather than a physical shift in the impingement limits.
- The predicted horn geometry varies substantially along the span, with changes in both horn size and orientation between the three spanwise stations.



ONERA M6 – Three-Dimensional Multilayer Ice Shape

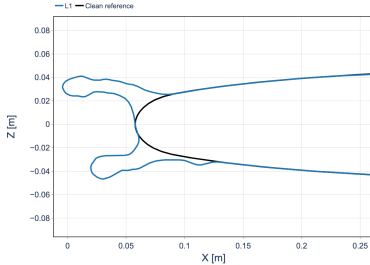
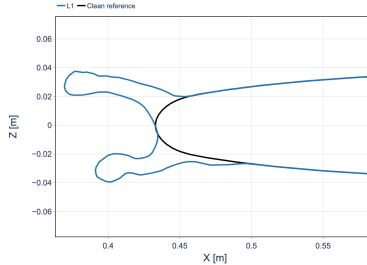
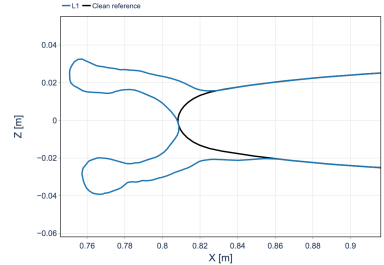


Multilayer Accretion

The level-set approach captures the three-dimensional evolution of the ice geometry and the spanwise development of the horn structures.



ONERA M6 – Multilayer Ice-Shape Comparison

 $y = 0.10 \text{ m}$  $y = 0.75 \text{ m}$  $y = 1.40 \text{ m}$ 

Multilayer Ice Shapes Assessment

- CHAMPS predicts a consistent multilayer ice morphology across the three spanwise locations.
- The horn size and extent vary along the span, reflecting the three-dimensional development of the accretion.



Conclusions

NACA0012

- CHAMPS predictions are generally well positioned within the IPW3 code-to-code envelope, with limited grid sensitivity.
- Eulerian and Lagrangian droplet formulations show good agreement in the predicted collection-efficiency distributions.
- Single-layer ice shapes capture the experimental horn orientation well, while multilayer simulations reproduce complex three-dimensional features.

ONERA M6

- CHAMPS predictions remain consistent with the overall IPW3 trends across the different physical quantities.
- Single-layer ice shapes remain consistent across the grid hierarchy, with L4 exhibiting the largest deviations due to its coarser surface discretization.
- Multilayer simulations capture a more pronounced horn formation compared with the single-layer prediction.

Overall

- CHAMPS demonstrates a consistent end-to-end icing capability, with the level-set framework capturing complex three-dimensional ice morphologies in multilayer simulations.
- Future work will focus on improving the roughness and ice-density models.



Acknowledgments

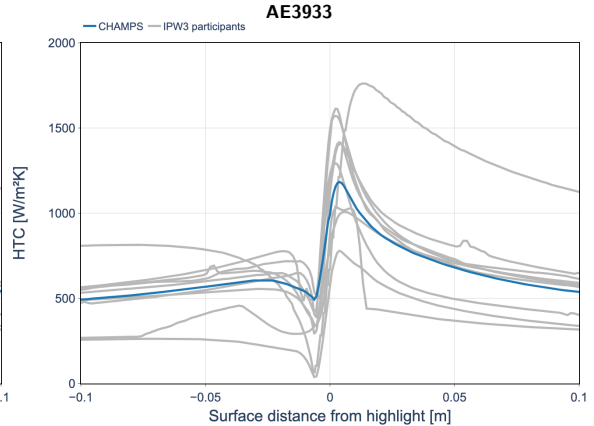
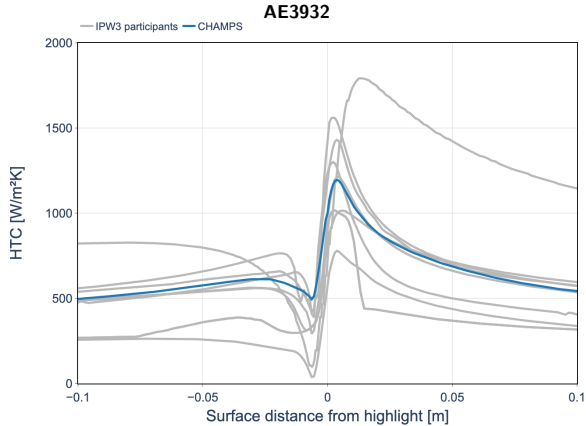
- This work is supported in part by a Bombardier/NSERC/CRIAQ Industrial Chair.
- Computations were performed on the Trillium supercomputer at the SciNet HPC Consortium.
- SciNet is funded by Innovation, Science and Economic Development Canada; the Digital Research Alliance of Canada; the Ontario Research Fund: Research Excellence; and the University of Toronto.



BACKUP SLIDES

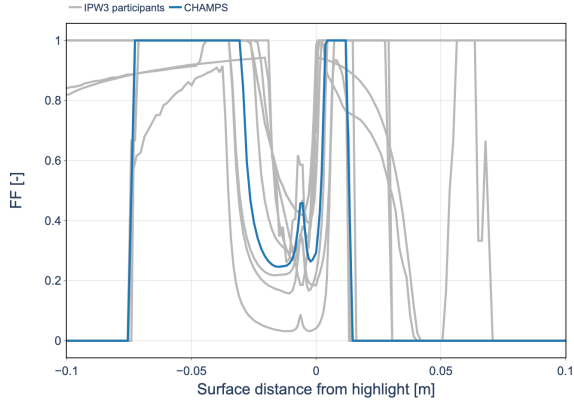


NACA0012 – Convective Heat-Transfer Coefficient

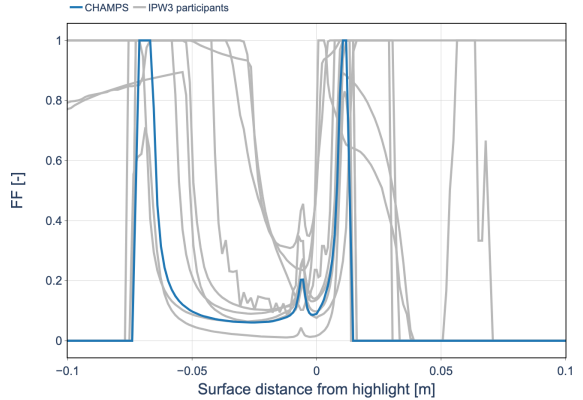


NACA0012 – Freezing Fraction

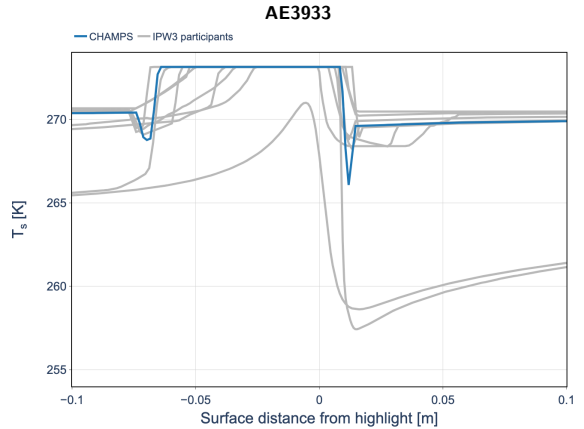
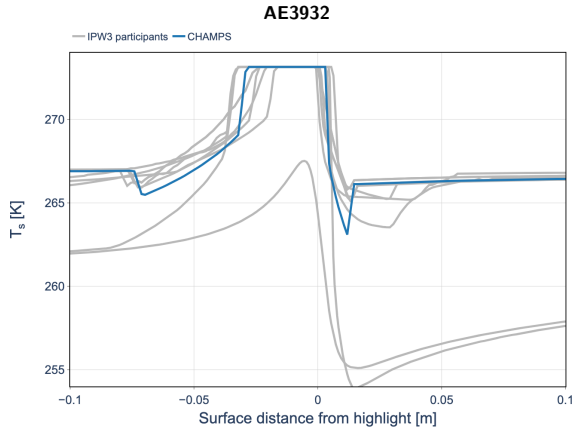
AE3932



AE3933

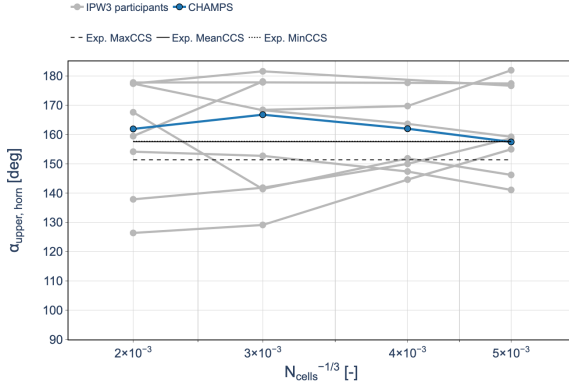


NACA0012 – Surface Temperature

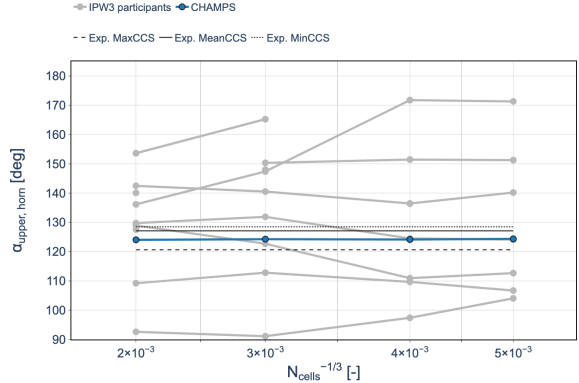


NACA0012 – Upper-Horn Angle

AE3932



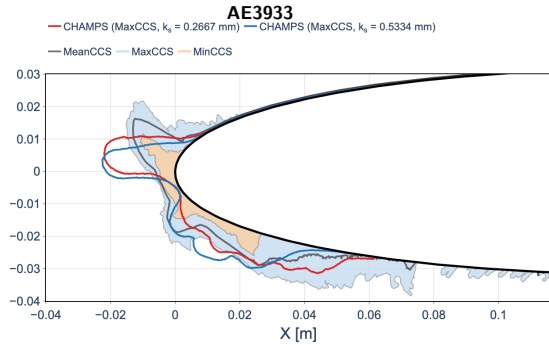
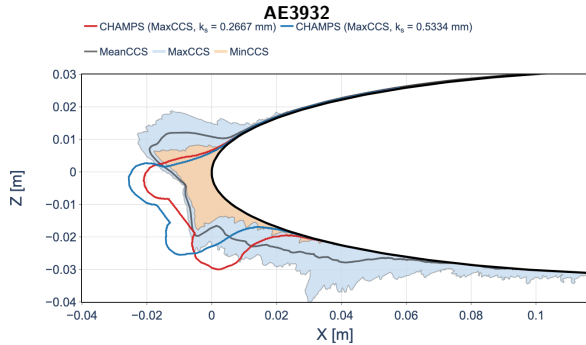
AE3933



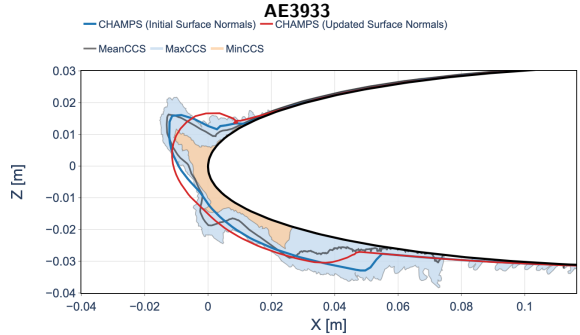
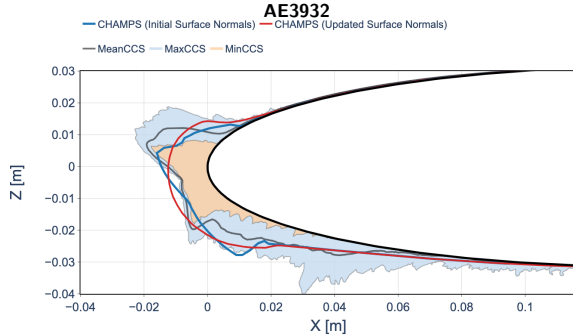
Assessment

- CHAMPS predicts upper-horn angles in very good agreement with the experimental measurements for both icing conditions.
- Droplet-bin refinement has negligible influence on the upper-horn angle (not shown due to limited variation).
- The substantial decrease in upper-horn angle from AE3932 to AE3933 is well captured.

NACA0012 – Multilayer Maximum Cross Sections



NACA0012 – Effect of Surface-Normal Update

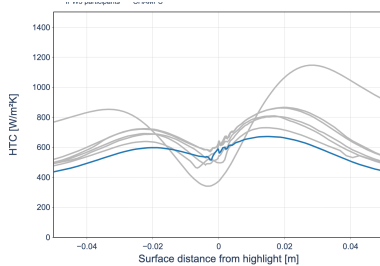
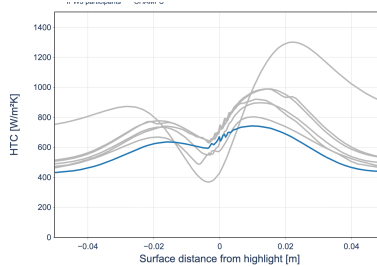
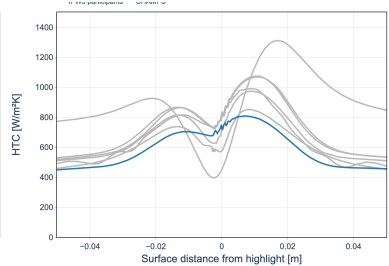


Assessment

- Updating the surface normals while retaining the original surface accretion map produces a blunter horn and modifies its orientation.
- Retaining the initial normals provides better agreement with the experimental horn orientation for the present algebraic displacement.
- The updating of the surface accretion solution with the evolving normals was not considered in the present approach and will be investigated further.

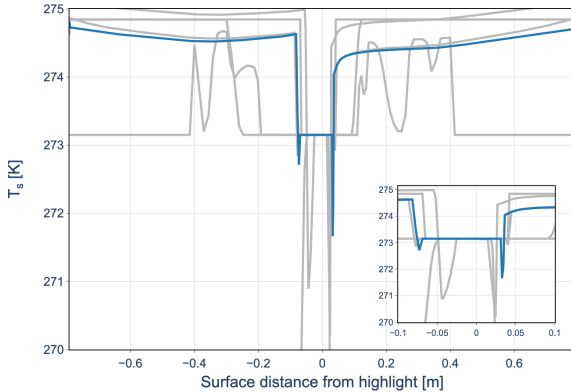


ONERA M6 – Convective Heat-Transfer Coefficient

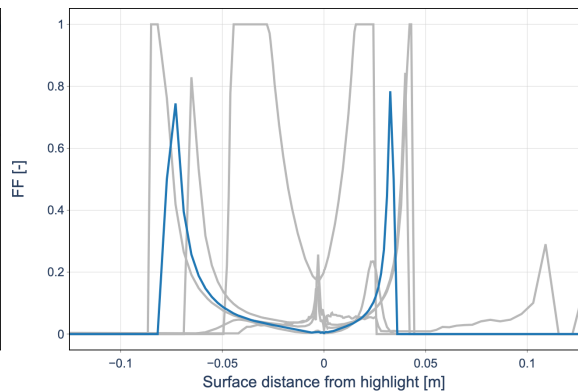
 $y = 0.10 \text{ m}$  $y = 0.75 \text{ m}$  $y = 1.40 \text{ m}$ 

ONERA M6 – Surface Thermodynamics ($y = 0.75$ m)

Surface Temperature

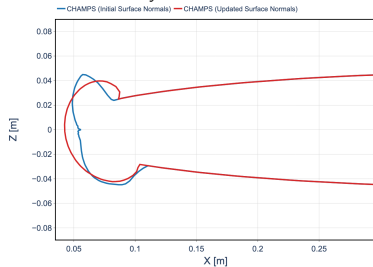


Freezing Fraction

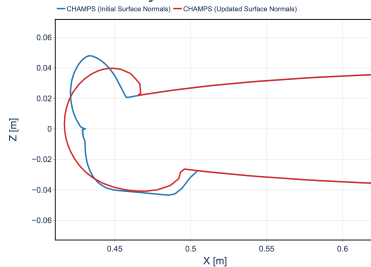


ONERA M6 – Effect of Surface-Normal Update

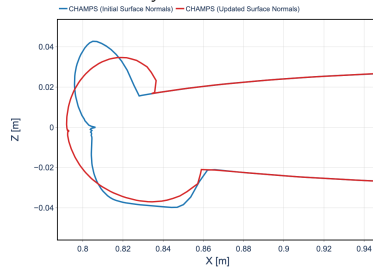
$y = 0.10 \text{ m}$



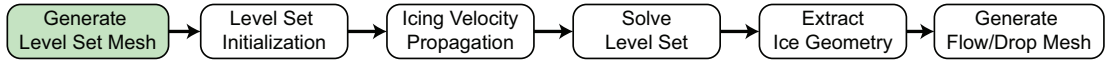
$y = 0.75 \text{ m}$



$y = 1.40 \text{ m}$

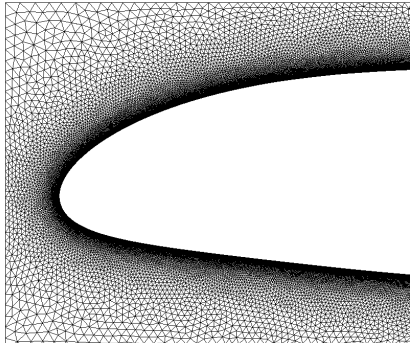


CHAMPS – Multilayer Level-Set Evolution

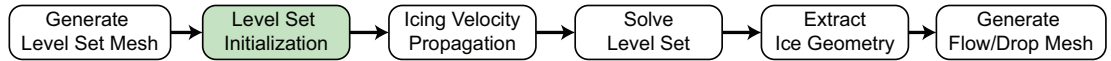


Generate Level-Set Mesh

A dedicated level-set mesh is generated to avoid the restrictive time-step requirements of the highly refined near-wall RANS mesh. The mesh is adapted to encompass the expected ice-thickness extent.

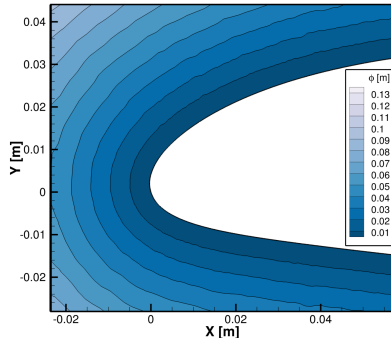


CHAMPS – Multilayer Level-Set Evolution

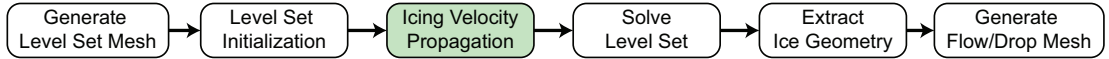


Level-Set Initialization

The level-set field is initialized as a signed-distance function from the current ice–air interface using an **Eikonal-based wall-distance method** or an equivalent wall-distance technique.



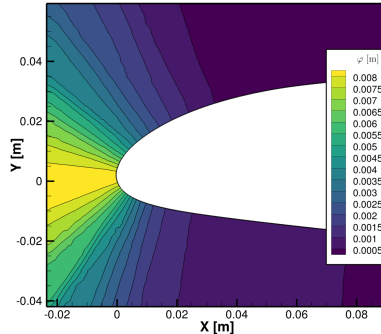
CHAMPS – Multilayer Level-Set Evolution



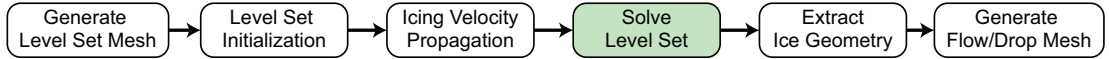
Icing Velocity Propagation

The interface velocity magnitude, $V_{ice} = h_{ice} / \Delta t_{layer}$, is propagated from the ice–air interface into the level-set domain. The interface velocity is then constructed using the level-set normal:

$$\mathbf{V} = V_{ice} \mathbf{n}_\phi.$$



CHAMPS – Multilayer Level-Set Evolution



Solve Level Set – Rosenbrock–Wanner (ROW) Scheme

- The level-set transport equation is advanced using a **two-stage Rosenbrock–Wanner (ROW) semi-implicit scheme**:

$$\phi^{n+1} = \phi^n + \sum_{j=1}^s m_j S_j, \quad \left(\frac{I}{\gamma \Delta t} + J \right)^n S_i = -R \left(\phi^n + \sum_{j=1}^{i-1} a_{ij} S_j \right).$$

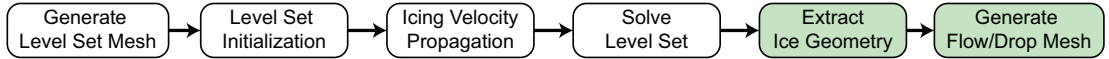
- The ROW scheme has demonstrated **excellent performance for flow time integration in CHAMPS**, providing improved stability and allowing larger time steps; it was therefore adopted for the level-set equation.
- At each ROW stage, a **linear system** is solved iteratively using **GMRES**:

$$\underbrace{\left(\frac{I}{\gamma \Delta t} + J \right)^n}_A S_i = -R \underbrace{\left(\phi^n + \sum_{j=1}^{i-1} a_{ij} S_j \right)}_{b_i}.$$

* ϕ^n : level-set solution; S_j : stage increment; s : number of ROW stages; m_j, a_{ij} : ROW coefficients; γ : ROW parameter; Δt : time step; I : identity matrix; $R(\phi)$: spatial residual; $J = \partial R / \partial \phi$: residual Jacobian.

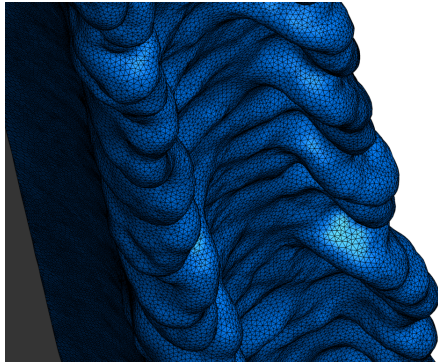


CHAMPS – Multilayer Level-Set Evolution



Extract Ice Geometry & Generate Volume Mesh

The updated ice surface is extracted from the volumetric level-set solution as the $\phi = 0$ isosurface using **MMG**. A new **volume mesh** is then generated around the updated ice geometry using **Pointwise Python scripts**.



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